

```

1 def string_sequence(n: int) -> str:
2     result = ''
3     for i in range(n + 1):
4         result += str(i) + ' '
5     return result[:-1]

```

(a) String Concatenation in Loops from HumanEval-15

```

1 def sort_matrix(M):
2     n = len(M)
3     for i in range(n):
4         for j in range(n-i-1):
5             if sum(M[j]) > sum(M[j+1]):
6                 M[j], M[j+1] = M[j+1], M[j]
7     return M

```

(b) Nested Looping from MBPP-12

```

1 for (Map.Entry<K, V> entry : map.entrySet()) {
2     V value = entry.getValue();
3     Set<K> keys = result.get(value);
4     if (keys == null) {
5         Keys = new HashSet<>();
6         result.put(value, keys);
7     }
8     keys.add(entry.getKey());
9 }

```

(c) Object Creation in Loops from AixBench-164